

Product Idea to be submitted to MSME Idea Hackathon 5.0

1. Title of proposed idea/innovation:



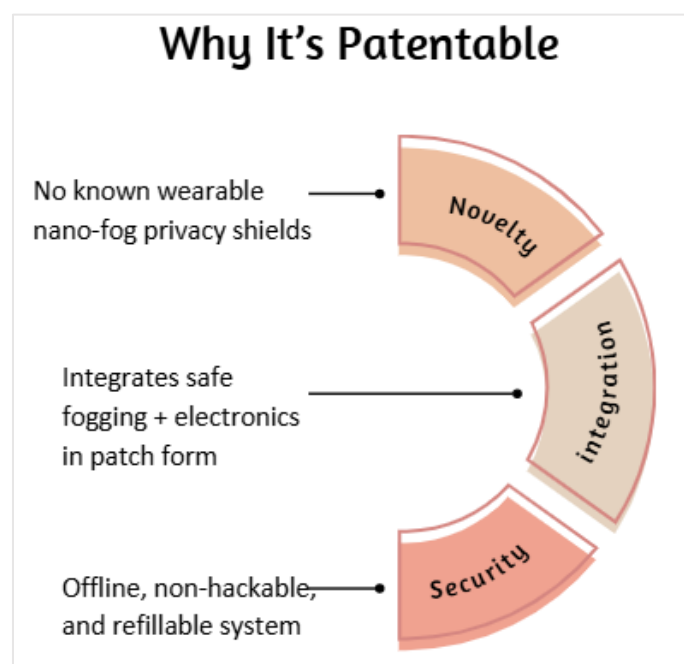
NANOSAFE CLOAK: A SMART, WEARABLE
NANO-FOG PRIVACY SHIELD
AGAINST HIDDEN CAMERA SURVEILLANCE

2. Use of Existing Intellectual Property

No, the core concept of the "NANOSAFE CLOAK: A Smart, Wearable Nano-Fog Privacy Shield Against Hidden Camera Surveillance" is an original and novel idea. However, the project may utilize or adapt certain existing technologies and open research related to:

- Nanomaterials (e.g., graphene oxide, TiO_2 , or SiO_2 NPs) for fogging or scattering light.
- Electro-optical privacy filters or smart glass principles (existing in displays).
- Wearable electronics frameworks for power management and activation (open-source hardware standards like Arduino or ESP32 may be referenced).

These technologies are used in different domains and are not patented for use in anti-surveillance wearable fog shields, making this a new IP opportunity. The integration and innovative application of these elements for privacy shielding using controlled nano-fog dispersion in wearable format represents a unique use-case that is eligible for patent filing upon prototyping.



2.1 Briefly explain newness/uniqueness of the innovation

The NANOSAFE CLOAK introduces a novel concept of a wearable nano-fog privacy shield that actively counters hidden camera surveillance. Unlike traditional passive solutions, this innovation uses smart-controlled nanoparticle dispersion (e.g., SiO_2 , TiO_2) to create a localized, dynamic fog that distorts camera vision without obstructing human sight. The wearable design ensures on-demand activation, offering mobility, discretion, and privacy in sensitive environments like trial rooms, hotel rooms, or public restrooms. Its interdisciplinary blend of nanotechnology, optoelectronics, and cybersecurity offers a first-of-its-kind solution for physical anti-surveillance, bridging the gap between personal safety and emerging covert threats.

2.2 Concept & Objective

The NANOSAFE CLOAK is a smart, wearable device designed to protect individuals from hidden camera surveillance by generating a nano-fog shield on demand. The core concept revolves around the integration of nano-engineered materials (such as SiO_2 , TiO_2 , or graphene oxide) that can disperse light-scattering particles when activated, creating a visual distortion barrier detectable to cameras but transparent to the human eye. This offers a portable and immediate privacy solution in vulnerable environments like trial rooms, restrooms, hotel spaces, or changing areas.

The primary objective is to develop a cost-effective, energy-efficient, and wearable cloak that provides physical and optical privacy, ensuring user safety against covert surveillance threats. By combining nanotechnology, smart textiles, and IoT-based activation, the innovation aims to empower individuals—particularly women and children—with control over their personal privacy in real-world settings. This project also aligns with MSME goals of tech-driven social impact and potential for scalable manufacturing.

2.3 The potential areas of application in industry/market

- Hospitality Industry
 - Privacy protection in hotel rooms, changing areas, and restrooms to prevent hidden camera spying.
- Retail & Fashion Industry
 - Security in trial rooms of clothing stores and shopping malls.
- Defense and Intelligence
 - Anti-surveillance gear for undercover agents and operatives in sensitive missions.
- Personal Safety & Consumer Wearables

- Everyday use for women, children, and travelers in public/private spaces prone to surveillance threats.
- Healthcare Sector
 - Patient privacy during telehealth consultations, diagnostic procedures, or in wearable hospital gowns.
- Event Management & Temporary Infrastructure
 - Temporary privacy installations in mobile restrooms, tents, and green rooms.
- Smart Homes and IoT Devices
 - Integration with smart mirrors, dressing areas, or voice-controlled privacy zones.
- Education & Hostels
 - Use in school or college dormitories and hostel bathrooms for added security.
- Automobile Interiors
 - Privacy protection inside cabs or ride-share vehicles against unauthorized dashcam recordings



Real-Time Analogy: The NanoSafe Cloak in Daily Life

From the moment you **check into a hotel room**, to trying on clothes at a **mall trial room**, or commuting in a **rideshare cab**, your privacy is constantly at risk.

Imagine wearing a **smart patch** that, with a simple tap, surrounds you in an **invisible fog shield**, blocking all hidden cameras instantly.

Whether you're a **traveler, agent, student, patient, or performer**, the NanoSafe Cloak adapts to your world — guarding your most vulnerable moments, silently.

2.4 Briefly provide the market data for the potential idea/ innovation

Segment	Market Size (India)	Growth Rate (CAGR)	Relevance to NANOSAFE CLOAK	Reference
Nanotechnology	USD 236 million (2024)	26.5%	Core to nano-fog dispersion mechanism	IMARC
Smart Wearables	USD 2.37 billion (2024)	16%	Form factor for wearable privacy device	IMARC
Smart Textiles	USD 160 million (2024)	20.5%	Potential material base for cloak design	IMARC
Surveillance Countermeasures	USD 1.2 billion (global, 2022)	9.8%	Direct application in anti-surveillance technology	Allied Market Research
MSME Sector – Tamil Nadu	4.95 million MSMEs (estimated)	--	Robust base for manufacturing, scaling, and local innovation	Wikipedia – TN MSME Dept.
Women Safety Devices Market	Projected USD 3.5 billion (by 2030)	9–12%	Key target demographic for privacy protection applications	Fortune Business Insights
Textile Industry – Tamil Nadu	~35% of India's textile production	--	Facilitates wearable integration and smart-fabric prototyping	IBEF
Defense/Surveillance Gear	Growing under “Make in India” Initiative	--	Opportunity for government procurement and specialized use cases	PIB India

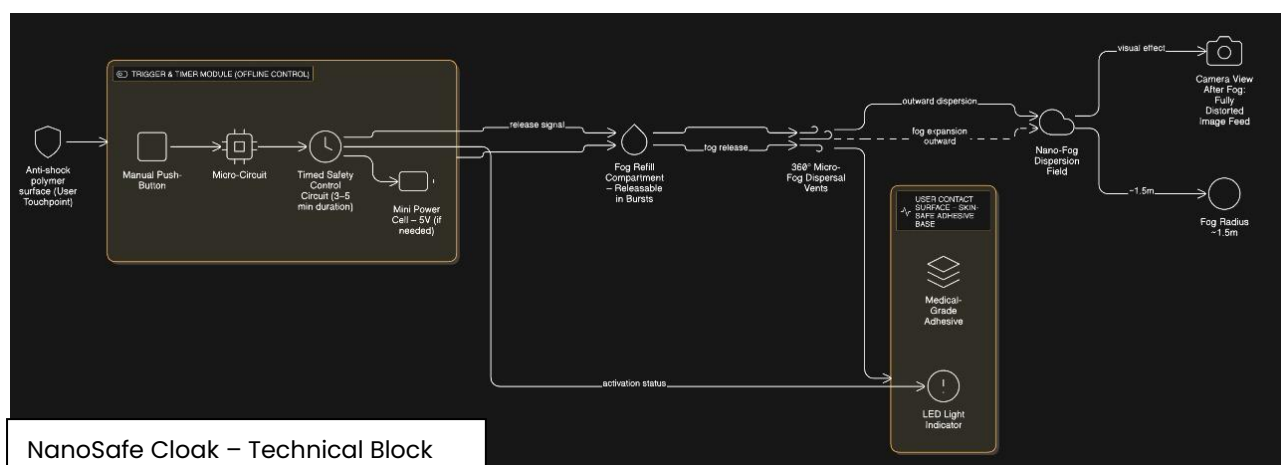
Market Readiness & Opportunities

- **Sector overlap:** Tamil Nadu's expertise in textiles and electronics complements integration of nanomaterials into wearable privacy tech.
- **Policy support:** The state's MSME department and bodies like TANSIDCO and EDII foster tech innovation, prototyping, and financing .
- **Rising privacy demand:** Growing awareness around hidden-camera threats in hospitality, retail, health, and urban living adds urgency.
- **Export potential:** Tamil Nadu's global textile position and India's push in smart textiles create export-ready pathways.

The convergence of rapid market growth in nanotechnology, smart wearables, and the strong industrial base of Tamil Nadu's MSMEs positions **NANOSAFE CLOAK** for high impact. Ongoing demand for personal privacy protection makes this innovation not only timely but also well-supported by regional ecosystems.

3. Technical Advancement Summary: A Novel Wearable Fog-Based Privacy Shield for Hidden Camera Protection

The NANOSAFE CLOAK is a smart, wearable privacy solution that uses nano-fog dispersion technology to counter hidden camera surveillance. It integrates nanomaterials (like TiO_2 or SiO_2) capable of creating a light-scattering fog layer when electrically activated, distorting camera capture while remaining visually clear to the human eye. Controlled via a compact electronic unit or mobile app, the system offers real-time, on-demand privacy. Unlike passive detection tools, this innovation provides active, physical anti-surveillance shielding in sensitive spaces. The key technical advancement lies in the novel integration of nanotechnology and wearable systems to deliver personalized optical cloaking in everyday environments.



4.Addressing Hidden Camera Surveillance in Private and Vulnerable Spaces

The main problem addressed by the NANOSAFE CLOAK is the growing threat of hidden camera surveillance in private and semi-private spaces such as trial rooms, restrooms, hotel rooms, hostels, and public toilets. With the miniaturization of cameras and their covert placement, individuals—especially women and children—face serious privacy violations and psychological distress. Existing solutions either rely on detection (which is not foolproof) or are not portable. This project aims to solve the lack of a reliable, wearable, and active anti-surveillance mechanism, offering a physical privacy shield that can be instantly activated to block hidden cameras in real-time.

5. Target Users and Beneficiaries

The NANOSAFE CLOAK is designed for individuals seeking protection from covert surveillance, especially:

- Women and children in public or private changing spaces.
- Travelers and hotel guests concerned about hidden cameras.
- Retail shoppers using trial rooms.
- Students in hostels, and individuals in shared accommodations.
- Undercover professionals such as journalists or intelligence agents.
- Privacy-conscious consumers in urban and smart-home environments.

6. Core Function and Purpose

The NANOSAFE CLOAK provides on-demand visual privacy by emitting a nano-fog layer that distorts the view of hidden cameras. When activated, it forms a light-scattering shield around the wearer, effectively blocking camera visibility without obstructing human vision. It will:

- Detect or be manually activated in suspected surveillance zones.
- Deploy a localized nano-fog using nanomaterials.
- Integrate with wearable electronics for smart control and portability.

7. Any unique features: Distinctive Features and Innovations

Yes, the NANOSAFE CLOAK has several unique features:

- Nano-Fog Privacy Shield: Uses safe, non-toxic nanomaterials to generate a fog that distorts camera sensors but not human vision.
- Wearable and Portable: Unlike stationary anti-surveillance devices, this is designed as a smart cloak, scarf, or shawl, allowing mobility and discretion.
- Smart Activation: Integrated with IoT-based controls or manual switches, enabling quick deployment in risky areas.
- Non-Invasive Solution: Provides privacy without detection or confrontation, making it ideal for real-world daily use.
- Customizable and Scalable: Can be adapted into various wearable forms (jackets, scarves, headgear), and scaled for mass production via MSMEs.

8(a). Implementation Complexity and Risk Assessment

- The implementation of the NanoSafe Cloak involves moderate complexity due to the integration of nanomaterials, wearable electronics, and fog dispersion technology. At its core, the cloak uses biodegradable, skin-safe nano-particles like SiO_2 or TiO_2 —already approved in cosmetic industries—combined with a compact sensor-controlled diffusion unit. From a fabrication standpoint, wearable integration and sensor design can follow existing IoT and textile manufacturing models.

Risks include:

- Ensuring uniform fog dispersion in all room types (especially with airflow like AC).
 - Avoiding particle inhalation (mitigated via FDA-compliant limits).
 - Preventing misuse (solved via biometric/manual triggers and use-limit caps).
 - Maintaining cost-efficiency and refilling mechanisms.
- Overall, the project is prototypable, scalable, and safe under controlled conditions, with risks being manageable through design and policy safeguards.

8.(b) Technological Landscape and Gaps in Personal Privacy Protection

With the miniaturization of electronics and the rise of covert surveillance, hidden cameras are increasingly deployed in restrooms, hotel rooms, and trial spaces—posing significant privacy and safety threats, especially to women and children. Traditional countermeasures like RF detectors, IR sensors, and UV flashlights require user expertise, offer inconsistent accuracy, and are reactive rather than protective. Research by Alù & Engheta (2005) on cloaking via metamaterials and light-scattering laid foundational work

for non-invasive visual distortion fields. Meanwhile, developments in nanomaterials like SiO₂ and TiO₂ have enabled safe, light-manipulating particles widely used in cosmetics and sunscreens. Products like anti-surveillance scarves or retro-reflective wear exist, but they are passive, ineffective under broad conditions, and not optimized for personal privacy-on-demand. Despite the market presence of these fragmented approaches, there is no integrated, wearable nano-fog technology offering **real-time, active protection**—which presents a compelling gap in the current technological landscape ripe for innovation.

1. **References:**

- Alù, A., & Engheta, N. (2005). Achieving transparency with plasmonic and metamaterial coatings. *Physical Review E*, 72(1).
- <https://www.nanowerk.com/news/newsid=41348.php> (nano-invisibility examples)

2. **Example:**

“According to a 2023 report, over 32% of women travelers reported concerns about hidden surveillance in accommodations.”

(Source: Fortune Business Insights, 2023)

3. **Briefly mention user concern / lack of awareness:**

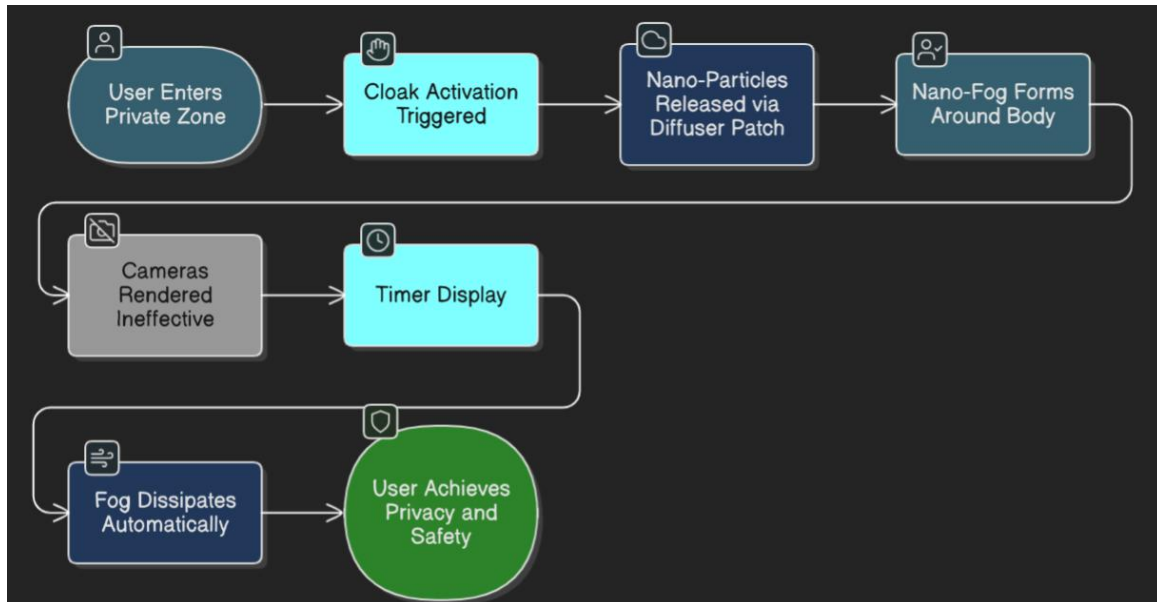
“Moreover, general public awareness of such threats is low, and few real-world solutions exist to empower end-users defensively rather than reactively.”

9.How is This Project Made and Used:

1. Imagine pressing a small button on your scarf, and a soft, invisible shield surrounds you—silently blurring the eyes of hidden cameras. That’s the **NanoSafe Cloak** in action.
2. Built using **biodegradable silica-based nanoparticles**, the cloak integrates a **micro-fog diffusion patch** powered by a **silent circuit (ESP32/MCU)** with **manual or biometric activation**. Once triggered, the fog disperses in a **controlled 2-meter dome**, scattering light and confusing camera sensors—while staying harmless and breathable to the human eye.
3. The cloak is **modular, refillable**, and runs **completely offline**, ensuring trust and safety. It’s everyday privacy—engineered like defense tech, disguised as wearable comfort.

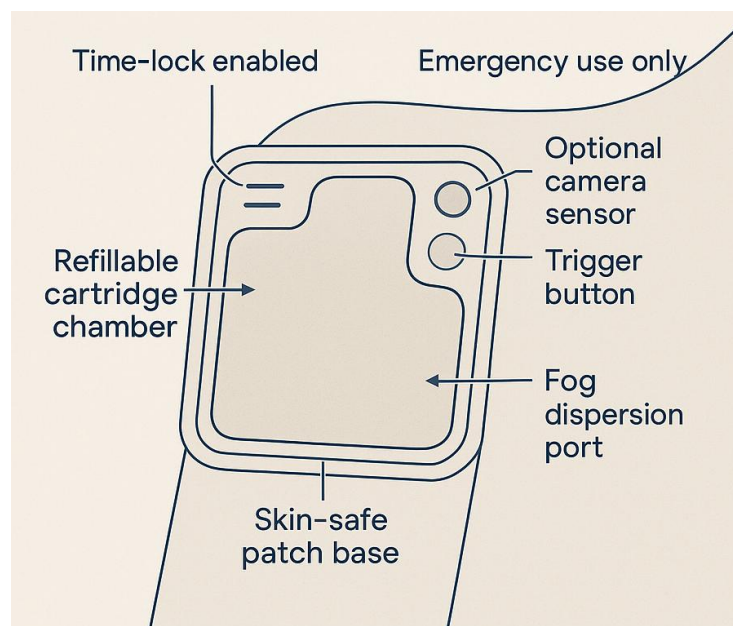
10. Block diagram/ flow chart/ Circuit Diagram

Flow chart:



"NanoSafe Cloak offers an active, wearable defense against hidden surveillance — empowering privacy at the push of a button."

"Built with smart nano-fog dispersion, it ensures real-time visual shielding without compromising user comfort or safety."



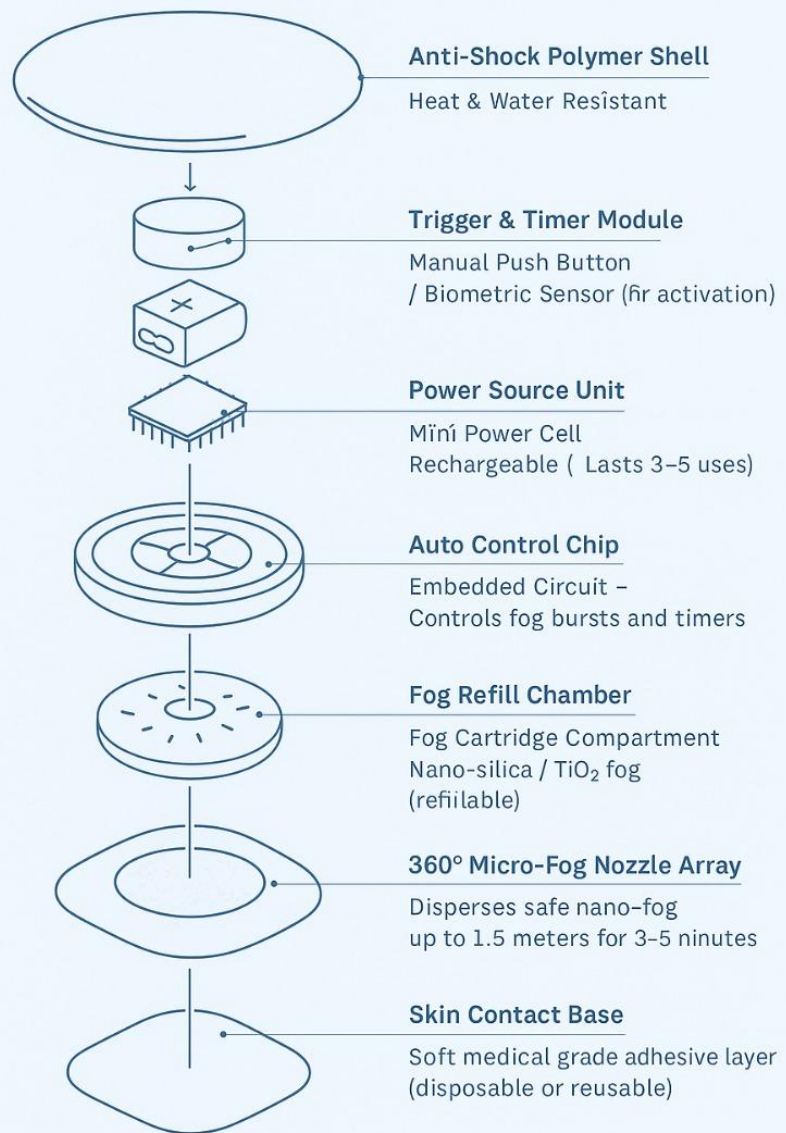


Figure: Internal Architecture of NanoSafe Cloak Patch Unit – Showing Core Modules for Privacy-Enabled Fog Dispersal

Fig 1. System Architecture Diagram(PROTOTYPE)

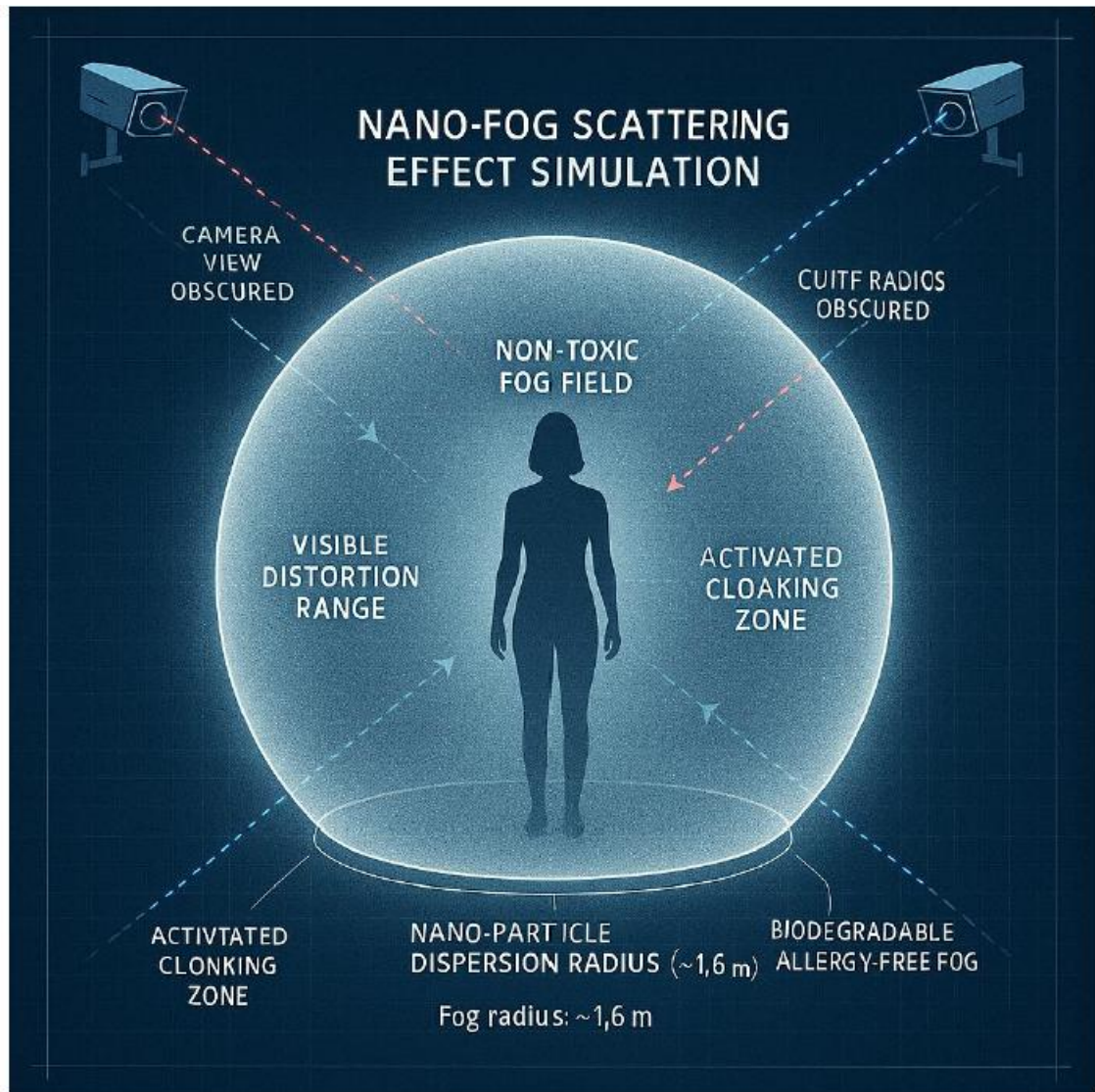


Fig 2. Fog Dispersion Simulation



Fig 3. Use Case Visual: “Smart Dust Shield – Use Case Workflow”



Fig 4. External Representation of Nano Safe Cloak



Visually see how our nano safe cloak works